

Notes: Autor, Levy & Murnane (QJE, 2003)
The Skill Content of Recent Technological Change: An Empirical Exploration

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Task data from the Dictionary of Occupational Titles	4
3.2	Employment data	4
3.3	Computerization measures	4
3.4	Education-demand translation	4
4	Model and empirical specifications	5
4.1	Task model assumptions	5
4.2	Task-input accounting	5
4.3	Industry computerization regressions	5
4.4	Within-occupation task changes	5
4.5	Demand for college-equivalent labor	6
5	Identification and empirical design	6
5.1	Identification logic	6
5.2	Why pre-trends matter	6
5.3	Why within-cell estimates matter	6
5.4	Threats and limitations	6
6	Tables and figures	7
6.1	Table I: Predictions of Task Model for the Impact of Computerization on Four Categories of Workplace Tasks; Figure I: Trends in Routine and Nonroutine Task Input, 1960 to 1998 . . .	7
6.2	Table II: Means of Task Input by Decade and Decomposition into Within and Between Industry Components; Figure II: Smoothed Differences between the Density of Task Input	8
6.3	Table III: Computerization and Industry Task Input, 1960-1998; Table IV: Computer Investment, Capital Intensity, and Task Input in Three-Digit Industries	9
6.4	Table V: Computerization and Industry Task Input 1980-1998: Overall and by Education Group; Table VI: Computerization and Changes in Job Task Content within Occupations 1977-1991	10
6.5	Table VII: Shifts in College-Equivalent Labor Demand Implied by Changes in Job Tasks, 1970-1998	11
7	Short summary	12
8	Conclusion	13
8.1	Core conclusion	13

8.2	Economic intuition	13
8.3	Incremental contribution revisited	13
8.4	Implications for wage inequality and polarization	13
8.5	Limitations	13
8.6	Takeaway	13

1 Big picture

- **Question.** What exactly is “skill-biased technical change”? Which tasks do computers substitute for, which tasks do they complement, and how does this reshape demand for education?
- **Core theory.** Computers substitute for workers in routine tasks that can be described by explicit rules. Computers complement workers in nonroutine cognitive tasks that require problem solving, judgment, and complex communication.
- **Task taxonomy.** The paper separates work into four categories: routine cognitive, routine manual, nonroutine analytic, and nonroutine interactive/manual activities.
- **Main prediction.** Industries and occupations that initially use many routine tasks adopt more computers as computer prices fall. They reduce labor input of routine tasks and raise labor input of nonroutine cognitive tasks.
- **Main empirical finding.** From 1960 to 1998, routine cognitive and routine manual task inputs fall sharply, while nonroutine analytic and interactive task inputs rise.
- **Within-cell evidence.** The patterns hold within industries, within occupations, and within education groups. This is important because the result is not just a shift from one occupation to another; the task content of nominally similar jobs changes.
- **Education implication.** Translating task shifts into skill demand, the model explains a large share of the observed increase in relative demand for college-equivalent labor between 1970 and 1998.
- **Economic takeaway.** Computers do not simply favor “skilled” workers. They substitute for routine task inputs and complement abstract, analytic, and interactive work. The skill demand shift is a task-content phenomenon.

2 Incremental contribution of the paper

- **From labels to mechanisms.** Earlier work documented that computer adoption correlated with rising demand for educated labor. This paper explains *why*: computers substitute for routine tasks and complement nonroutine cognitive tasks.
- **Task-based theory of technical change.** The paper moves the literature from a two-skill model of high-skill versus low-skill labor to a task model that separates what workers actually do at work.
- **New measurement strategy.** It links Dictionary of Occupational Titles task measures to Census and CPS employment data from 1960 to 1998, creating long-run task-input series for the U.S. economy.
- **Within-occupation evidence.** It shows that almost half of the effect operates through changes in task content within nominally identical occupations, not only through employment shifts across occupations.
- **Computerization as a task-demand shifter.** The paper connects task changes to industry computer investment and worker computer use, making the mechanism empirically operational.
- **Education-demand accounting.** It translates predicted task shifts into college-equivalent labor demand and shows that the task model can explain much of the observed rise in college demand.
- **Foundation for later “routine-biased technological change.”** The paper provides the canonical empirical and conceptual foundation for later work on job polarization, automation, and routine task displacement.

Interpretation: The incremental contribution is not merely documenting that technology is skill-biased. It redefines skill bias as task bias: the economically relevant distinction is whether a task can be codified and automated, not simply whether the worker has more schooling.

3 Data and measurement

3.1 Task data from the Dictionary of Occupational Titles

The paper uses the Dictionary of Occupational Titles (DOT) to assign occupational task measures. The five main task inputs are:

- nonroutine analytic;
- nonroutine interactive;
- routine cognitive;
- routine manual;
- nonroutine manual.

These DOT task measures are linked to employment by occupation, gender, education, industry, and year using Census and Current Population Survey data.

Interpretation: DOT task scores are occupation-level measures, but the paper converts them into economywide task inputs by weighting each occupation’s task content by its employment share in each year or cell.

3.2 Employment data

The main employment samples come from:

- Census IPUMS for 1960, 1970, and 1980;
- CPS Merged Outgoing Rotation Group samples for 1980, 1990, and 1998.

The analysis constructs industry-gender-education cells and occupation-level measures to track both extensive-margin and intensive-margin task changes.

Interpretation: The long time span matters. The paper compares pre-computer changes in the 1960s with changes after computer adoption accelerates in the 1970s, 1980s, and 1990s.

3.3 Computerization measures

The paper uses two main measures of computerization:

- NIPA industry computer capital investment per worker;
- CPS worker reports of computer use at work in 1984 and 1997.

Interpretation: The NIPA measure captures industry-level computer investment. The CPS measure captures direct workplace computer use. The fact that both tell similar stories strengthens the task-based interpretation.

3.4 Education-demand translation

The paper maps task changes into demand shifts for college-equivalent relative to noncollege-equivalent labor. Conceptually:

$$\Delta \log \left(\frac{D_C}{D_N} \right) = \text{task-price weights} \times \Delta \text{task inputs}.$$

The authors compare this task-implied demand shift to a standard constant-elasticity-of-substitution estimate of rising demand for college labor.

Interpretation: The education translation is a bridge from task evidence to the wage-inequality literature. It asks how much of the observed college demand shift can be explained by the observed changes in job tasks.

4 Model and empirical specifications

4.1 Task model assumptions

The paper builds on three assumptions:

1. computer capital is more substitutable for human labor in routine tasks than in nonroutine tasks;
2. routine and nonroutine tasks are imperfect substitutes;
3. greater intensity of routine inputs raises the marginal productivity of nonroutine inputs.

Interpretation: These assumptions imply that cheaper computers reduce labor demand in routine tasks and increase demand for nonroutine analytic and interactive tasks.

4.2 Task-input accounting

Economywide task input for task k in year t is measured as an employment-weighted average:

$$Task_{k,t} = \sum_o s_{o,t} DOT_{o,k},$$

where $s_{o,t}$ is occupation o 's employment share and $DOT_{o,k}$ is its task score.

Interpretation: The key empirical object is not wage or education directly. It is the evolution of task input in the economy.

4.3 Industry computerization regressions

The paper estimates regressions of changes in task input on changes in computerization:

$$\Delta Task_{j,t}^k = \alpha_t + \beta_k \Delta ComputerUse_{j,t} + \varepsilon_{j,t}^k,$$

where j indexes industries or occupations and k indexes task types.

Interpretation: The predicted signs are $\beta_k > 0$ for nonroutine analytic and interactive tasks, and $\beta_k < 0$ for routine cognitive and routine manual tasks.

4.4 Within-occupation task changes

To test whether task content changes within jobs, the paper estimates models linking worker computer use in occupations to changing task input within the same occupation:

$$\Delta Task_{o,t}^k = \alpha + \beta_k ComputerUse_{o,1984/1997} + \Gamma' X_{o,t} + u_{o,t}^k.$$

Interpretation: This is critical because a pure between-occupation story would say workers simply move from routine occupations to nonroutine occupations. The paper shows that task use shifts even within occupational titles.

4.5 Demand for college-equivalent labor

The paper converts task shifts into implied changes in relative demand for college-equivalent labor using fixed coefficients of education task input and standard elasticity assumptions. It then compares the task-based estimates with conventional CES estimates of relative demand.

Interpretation: This final step asks whether the task model can account for wage-inequality facts, not only task-content facts.

5 Identification and empirical design

5.1 Identification logic

This is not a randomized experiment. The identification is a model-guided empirical design. The paper tests whether changes in task composition line up with predictions from a task-based model of computerization.

- The first source of variation is time: computer prices fall sharply and computer use rises after the 1970s.
- The second source is cross-industry and cross-occupation intensity: some sectors adopt computers more rapidly than others.
- The third source is task composition: initially routine-intensive occupations should change more strongly.

Interpretation: The credibility comes from multiple aligned predictions: routine tasks fall, nonroutine cognitive tasks rise, effects are strongest in computerizing sectors, and changes occur within industries, occupations, and education groups.

5.2 Why pre-trends matter

The paper compares the 1960s with later decades. If task changes were simply a pre-existing trend unrelated to computers, routine and nonroutine task trends should look similar before and after computer diffusion. Instead, routine cognitive and routine manual tasks reverse earlier patterns, while nonroutine analytic and interactive tasks accelerate.

Interpretation: The 1960s serve as a useful benchmark for task trends before widespread computer adoption.

5.3 Why within-cell estimates matter

The paper decomposes aggregate task change into within-industry, between-industry, within-occupation, and between-occupation components. It also shows patterns within education groups.

Interpretation: This reduces the concern that the results are only re-labeling compositional shifts across broad sectors or education categories. The content of work changes inside narrowly defined cells.

5.4 Threats and limitations

- **Measurement of tasks.** DOT task measures are occupation-level proxies and may not capture all within-occupation changes.
- **Endogeneity of computer adoption.** Industries may computerize because demand for certain tasks was already changing.

- **Other technologies.** Organizational change, trade, and non-computer technologies may also alter task demand.
- **Education mapping.** The conversion from tasks to college demand relies on fixed coefficients and elasticity assumptions.
- **No firm-level causal shock.** The paper is a broad empirical exploration, not a micro causal design with randomized computer adoption.

Interpretation: The strength of the paper is not a single causal coefficient. It is the consistency between a theoretically precise task model and many independent empirical patterns.

6 Tables and figures

6.1 Table I: Predictions of Task Model for the Impact of Computerization on Four Categories of Workplace Tasks; Figure I: Trends in Routine and Nonroutine Task Input, 1960 to 1998

Read:

- Table I classifies routine versus nonroutine tasks and analytic/interactive versus manual tasks.
- Computers are predicted to substantially substitute for routine cognitive and routine manual tasks.
- Computers are predicted to strongly complement nonroutine analytic and interactive tasks.
- Figure I shows the central long-run fact: nonroutine analytic and interactive tasks rise, while routine cognitive and routine manual tasks fall after the 1970s.

Economic message: The theory in Table I directly predicts the empirical pattern in Figure I. The sharp post-1970 divergence is the paper’s visual summary of routine-biased technical change. The figure shows the task content of the economy moving away from codifiable work and toward problem-solving and communication work. This is a task interpretation of rising demand for educated labor.

TABLE I PREDICTIONS OF TASK MODEL FOR THE IMPACT OF COMPUTERIZATION ON FOUR CATEGORIES OF WORKPLACE TASKS		
	Routine tasks	Nonroutine tasks
	Analytic and interactive tasks	
Examples	- Record-keeping - Calculation - Repetitive customer service (e.g., bank teller)	- Forming/testing hypotheses - Medical diagnosis - Legal writing - Persuading/selling - Managing others
Computer impact	Substantial substitution	Strong complementarities
	Manual tasks	
Examples	- Picking or sorting - Repetitive assembly	- Janitorial services - Truck driving
Computer impact	Substantial substitution	Limited opportunities for substitution or complementarity

(a) Table I: task taxonomy and predicted computer impacts.

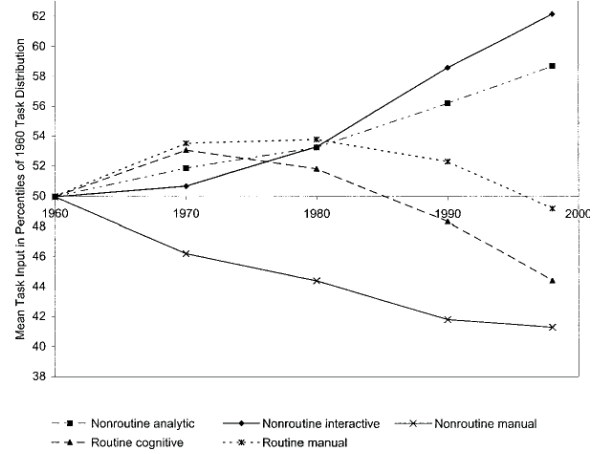


FIGURE I
Trends in Routine and Nonroutine Task Input, 1960 to 1998

Figure I is constructed using *Dictionary of Occupational Titles* [1977] task measures by gender and occupation paired to employment data for 1960 and 1970 Census and 1980, 1990, and 1998 Current Population Survey (CPS) sample Data are aggregated to 1120 industry-gender-education cells by year, and each cell is assigned a value corresponding to its rank in the 1960 distribution of task input (calculated across the 1120, 1960 task cells). Plotted values depict the employment-weighted mean of each assigned percentile in the indicated year. See Table I and Appendix 1 for definitions and examples of task variables.

(b) Figure I: long-run trends in task input.

Theory and first fact. Table I gives the four-task framework; Figure I shows that the predicted task shifts occur in the aggregate data.

6.2 Table II: Means of Task Input by Decade and Decomposition into Within and Between Industry Components; Figure II: Smoothed Differences between the Density of Task Input

Read:

- Table II decomposes changes in task inputs into within- and between-industry components.
- Figure II shows distributional changes across the 1960 task percentile distribution.
- Nonroutine analytic and interactive inputs grow especially in the upper part of the task distribution.
- Routine cognitive and routine manual inputs decline, especially after the 1970s.

Economic message: The task shift is not just an average change. The entire distribution of employment across task intensity changes, and much of the change occurs within industries. The figure makes clear that high-task-intensity occupations in nonroutine domains expand, while employment density in routine domains contracts. The table quantifies the same movement by decade and industry margin.

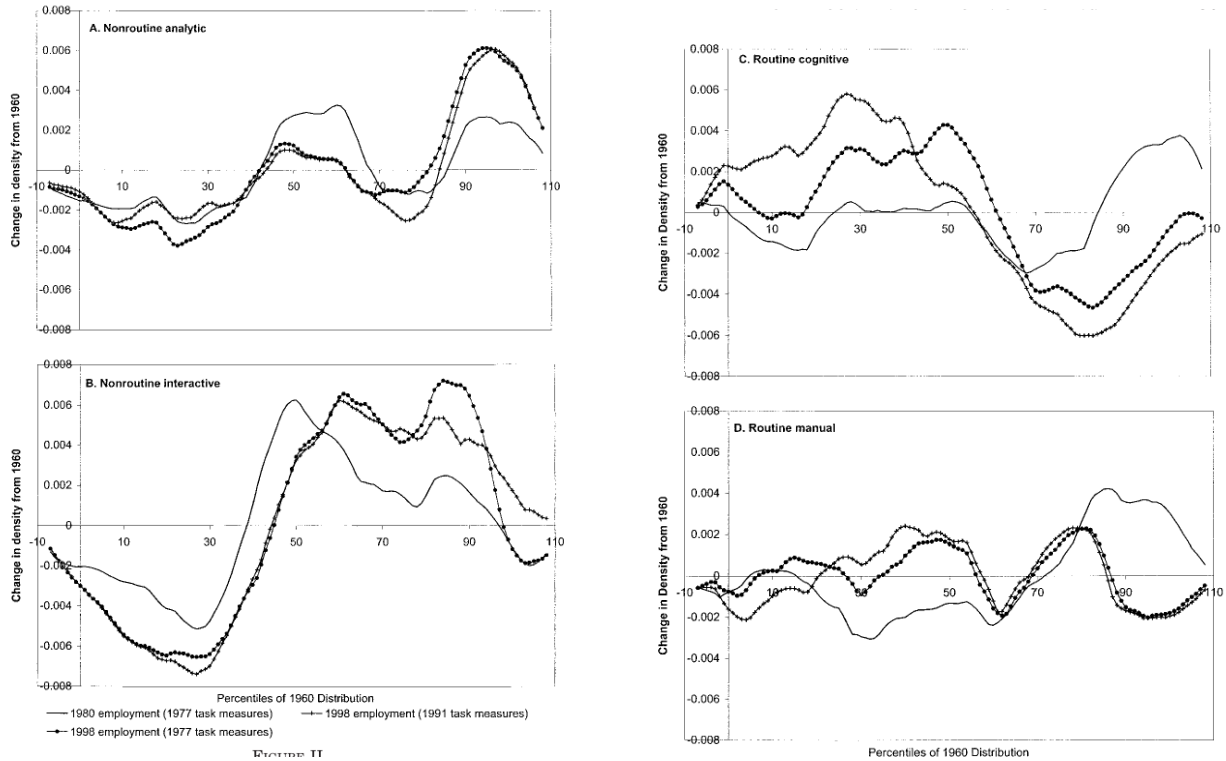
TABLE II
MEANS OF TASK INPUT BY DECADE AND DECOMPOSITION INTO WITHIN AND BETWEEN INDUSTRY COMPONENTS, 1960–1998

	A. Weighted means of economywide task input by decade (percentiles of 1960 task distribution)														
	1. Nonroutine analytic			2. Nonroutine interactive			3. Routine cognitive			4. Routine manual			5. Nonroutine manual		
	All	Male	Fem	All	Male	Fem	All	Male	Fem	All	Male	Fem	All	Male	Fem
1960 (Census)	50.0	54.6	37.0	50.0	58.4	26.4	50.0	48.6	53.8	50.0	43.2	69.2	50.0	56.5	31.6
1970 (Census)	52.5	57.5	41.5	51.1	61.7	27.5	52.2	49.8	57.5	54.0	44.9	74.4	46.9	54.1	30.8
1970 (Census)	51.9	56.4	41.8	50.7	60.8	28.1	53.1	50.8	58.2	53.5	44.7	73.3	46.2	53.2	30.7
1980 (Census)	54.9	58.2	49.2	55.4	63.3	41.8	52.9	50.0	58.0	55.1	44.6	73.3	44.0	53.2	28.1
1980 (CPS)	53.2	56.6	47.9	53.3	61.4	40.4	51.8	50.0	54.8	53.8	43.3	70.4	44.4	55.0	27.5
1990 (CPS)	56.2	57.4	54.6	58.6	62.7	53.0	48.3	48.1	48.6	52.3	42.8	65.5	41.8	53.1	26.3
1998 (CPS)	58.7	59.3	58.0	62.2	63.9	59.9	44.4	46.6	41.6	49.2	41.5	59.3	41.3	52.6	26.5

	B. Decomposition of task shifts into between and within industry components for combined genders (10 × annual changes in mean task percentile)											
	Total	Btwn	Withn	Total	Btwn	Withn	Total	Btwn	Withn	Total	Btwn	Withn
1960–1970	2.57	1.74	0.83	1.15	−0.34	1.49	2.20	1.14	1.06	4.01	2.39	1.62
1970–1980	3.02	1.54	1.48	4.88	0.26	4.42	−0.14	0.33	−0.47	1.63	0.79	0.84
1980–1990	2.97	0.92	2.05	5.31	0.52	4.79	−3.48	−1.42	−2.07	−1.47	−0.16	−1.31
1990–1998	3.12	0.67	2.45	4.48	0.54	3.94	−4.88	−1.31	−3.57	−3.88	−0.38	−3.50

Sources: Dictionary of Occupational Titles (1977), and all employed workers ages 18–64, Census IPUMS 1960, 1970, 1980, CPS MORG 1980, 1990, and 1998. Samples used for decadal changes in panel B are 1960–1970, 1960 and 1970 Census, 1970–1980, 1970 and 1980 Census, 1980–1990, 1980 and 1990 CPS MORG, 1990–1998, 1990 and 1998 CPS MORG. Two Census 1970 samples are used in panels A and B, one coded for consistency with the 1960 Census occupation codes and a second coded for consistency with the 1980 Census occupation codes. Data are aggregated to 1120 industry-gender-education cells by year and each cell is assigned a value corresponding to its rank in the 1960 distribution of task input calculated across the 1120, 1960 task cells. Panel A contains the employment-weighted means of each assigned percentile in the indicated year. Panel B presents a decomposition of the aggregate change in task input over the indicated years into within and between industry components for 140 consistent Census Industry Code (CIC) industries (59 in manufacturing, 81 in nonmanufacturing). See Table I and Appendix I for definitions and examples of task variables.

Table II: Means of task input and within/between-industry decompositions. The rotated table reports decade-level task means and decompositions by industry margins.



(a) Figure II, panels A-B: nonroutine analytic and interactive density changes.

(b) Figure II, panels C-D: routine cognitive and manual density changes.

Figure II: Smoothed differences in task-density distributions. The nonroutine panels show density gains in higher task percentiles; the routine panels show shifts away from routine task input.

6.3 Table III: Computerization and Industry Task Input, 1960-1998; Table IV: Computer Investment, Capital Intensity, and Task Input in Three-Digit Industries

Read:

- Table III relates changes in industry task input to growth in computer use.
- Computerization predicts growth in nonroutine analytic and interactive tasks.
- Computerization predicts declines in routine cognitive and routine manual tasks.
- Table IV shows similar patterns using NIPA computer investment and controls for capital intensity.

Economic message: The model predicts that computerizing industries should reorganize work away from routine tasks and toward nonroutine cognitive tasks. Tables III and IV provide direct industry-level evidence. The signs of the coefficients are the key reading rule. Positive for nonroutine analytic/interactive and negative for routine cognitive/manual is exactly the model’s prediction.

TABLE III
COMPUTERIZATION AND INDUSTRY TASK INPUT, 1960–1998
DEPENDENT VARIABLE: $10 \times$ ANNUAL WITHIN-INDUSTRY CHANGE IN TASK INPUT,
MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

		1. 1990– 1998	2. 1980– 1990	3. 1970– 1980	4. 1960– 1970
A. Δ Nonroutine analytic	Δ Computer use	12.04	14.02	9.11	7.49
	1984–1997	(4.74)	(4.97)	(4.17)	(5.28)
	Intercept	0.07	−0.66	−0.26	−0.55
		(1.00)	(1.03)	(0.86)	(1.05)
	R^2	0.04	0.05	0.03	0.01
B. Δ Nonroutine interactive	Weighted mean Δ	2.45	2.05	1.48	0.83
	Δ Computer use	14.78	17.21	10.81	7.55
	1984–1997	(5.48)	(6.32)	(5.71)	(6.64)
	Intercept	1.02	1.46	2.35	0.10
		(1.15)	(1.31)	(1.17)	(1.32)
C. Δ Routine cognitive	R^2	0.05	0.05	0.03	0.01
	Weighted mean Δ	3.94	4.79	4.42	1.49
	Δ Computer use	−17.57	−13.94	−11.00	−3.90
	1984–1997	(5.54)	(5.72)	(5.40)	(4.48)
	Intercept	−0.11	0.63	1.63	1.78
D. Δ Routine manual		(1.17)	(1.19)	(1.11)	(0.89)
	R^2	0.07	0.04	0.03	0.01
	Weighted mean Δ	−3.57	−2.07	−0.47	1.06
	Δ Computer use	−24.72	−5.94	−6.56	4.15
	1984–1997	(5.77)	(5.64)	(4.84)	(3.50)
	Intercept	1.38	−0.16	2.09	0.85
		(1.22)	(1.17)	(0.99)	(0.70)
	R^2	0.12	0.01	0.01	0.01
	Weighted mean Δ	−3.50	−1.31	0.84	1.62

n is 140 consistent CIC industries. Standard errors are in parentheses. Each column of panels A–D presents a separate OLS regression of ten times the annual change in industry-level task input between the endpoints of the indicated time interval (measured in centiles of the 1960 task distribution) on the annual percentage point change in industry computer use during 1984–1997 (mean 0.193) and a constant. Computer use is the fraction of industry workers using a computer at their jobs, estimated from the October 1984 and 1997 CPS samples. Estimates are weighted by mean industry share of total employment in FTEs over the endpoints of the years used to form the dependent variable. Samples used are Census 1960, 1970, and 1980 and CPS MORG 1980, 1990, and 1998. See Table I and Appendix 1 for definitions and examples of task variables.

(a) Table III: industry computerization and task-input changes.

Computerization regressions. Both tables connect task shifts to computerization, using different computer measures and regression structures.

TABLE IV
COMPUTER INVESTMENT, CAPITAL INTENSITY, AND TASK INPUT IN THREE-DIGIT
INDUSTRIES 1960–1998: STACKED FIRST-DIFFERENCE ESTIMATES
DEPENDENT VARIABLE: $10 \times$ ANNUAL CHANGE IN QUANTILES OF TASK MEASURE,
MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

	A. Δ Nonroutine analytic		B. Δ Nonroutine interactive		C. Δ Routine cognitive		D. Δ Routine manual	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
	Log(CI/L)		Log(KI/L)		Δ Log(KI/L)		1970–1980 dummy	
	6.65	6.76	11.59	10.03	−8.27	−8.30	−9.11	−8.20
	(4.13)	(3.97)	(3.21)	(3.31)	(3.63)	(3.26)	(2.57)	(2.29)
	[6.36]	[5.90]	[3.97]	[4.50]	[4.74]	[3.76]	[3.27]	[2.86]
	1.22		−3.41		−2.93		−2.42	
	(4.36)		(4.58)		(4.76)		(3.95)	
	[6.36]		[4.45]		[7.31]		[5.87]	
		0.24		3.01		−1.32		−3.89
		(2.35)		(2.24)		(2.12)		(1.92)
		[2.36]		[2.19]		[2.18]		[2.38]
	−0.64	−0.54	1.38	2.49	−0.32	−0.84	0.68	−0.80
	(1.08)	(1.29)	(1.50)	(1.62)	(1.31)	(1.58)	(0.96)	(1.17)
	[1.02]	[1.23]	[2.07]	[2.12]	[1.13]	[0.93]	[0.94]	[1.03]
	−0.34	−0.25	0.58	1.83	−1.62	−2.14	−1.32	−2.90
	(1.57)	(1.60)	(1.81)	(1.70)	(1.56)	(1.86)	(1.11)	(1.38)
	[1.43]	[1.67]	[1.58]	[1.46]	[1.33]	[1.35]	[0.86]	[1.07]
	−1.19	−1.13	−1.91	−0.90	−1.33	−1.71	−1.15	−2.36
	(1.55)	(1.62)	(1.83)	(1.70)	(1.66)	(1.64)	(1.32)	(1.47)
	[1.77]	[1.93]	[1.85]	[1.88]	[1.93]	[1.63]	[0.95]	[1.13]
	8.89	8.23	12.40	11.30	−9.29	−7.09	−9.62	−5.55
	(4.08)	(4.42)	(4.25)	(3.59)	(4.12)	(3.63)	(3.14)	(2.67)
	[5.45]	[6.38]	[4.76]	[4.80]	[4.48]	[4.36]	[3.75]	[3.30]
	R^2	0.06	0.06	0.11	0.12	0.14	0.14	0.20
Weighted mean of dependent variable								
1960–1970	1.16		1.74		1.30		1.63	
1970–1980	1.23		4.59		−0.20		0.98	
1980–1990	2.07		4.69		−2.05		−1.74	
1990–1998	2.15		3.76		−3.03		−2.82	

$n = 492$. Robust standard errors in parentheses are heteroskedasticity consistent and account for clustering of errors within 42 consistent NIPA sectors in each decade (168 clusters). Standard errors in brackets additionally account for potential serial correlation within sectors (42 clusters). Each column presents a separate OLS regression of ten times annual industry changes in task input on the indicated

(b) Table IV: computer investment, capital intensity, and task input.

6.4 Table V: Computerization and Industry Task Input 1980–1998: Overall and by Education Group; Table VI: Computerization and Changes in Job Task Content within Occupations 1977–1991

Read:

- Table V shows that computerization-task relationships appear within education groups, not only between education groups.
- The within-education effects are strongest for high school graduates and workers with some college.
- Table VI shows that occupations with more computer use experience within-occupation task changes.
- This means nominally identical jobs change their task content over time.

Economic message: The task story is deeper than a shift from low-education to high-education employment. The tasks performed by workers with the same education and workers in the same occupation also change. These tables are central for the paper’s contribution. They show that the computerization effect is not a purely compositional reallocation across broad worker types; technology changes what jobs require.

TABLE V
COMPUTERIZATION AND INDUSTRY TASK INPUT 1980–1998:
OVERALL AND BY EDUCATION GROUP
DEPENDENT VARIABLE: $10 \times$ ANNUAL CHANGE IN QUANTILES OF TASK MEASURE,
MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

	1. Δ Nonroutine analytic	2. Δ Nonroutine interactive	3. Δ Routine cognitive	4. Δ Routine manual
A. Aggregate within-industry change				
Δ Computer use	12.95	15.97	-15.84	-14.32
1984–1997	(3.68)	(4.32)	(4.73)	(4.73)
Intercept	-0.33	1.27	0.38	0.54
	(0.77)	(0.90)	(0.99)	(0.99)
Weighted mean task Δ	2.20	4.39	-2.71	-2.25
B. Within industry: High school dropouts				
Δ Computer use	4.64	11.92	-2.64	-8.85
1984–1997	(6.07)	(8.73)	(7.95)	(6.76)
Intercept	-2.51	-4.39	0.02	1.11
	(1.26)	(1.82)	(1.66)	(1.41)
Weighted mean task Δ	-1.61	-2.07	-0.49	-0.62
C. Within industry: High school graduates				
Δ Computer use	0.04	13.49	-28.18	-25.50
1984–1997	(4.17)	(5.40)	(6.13)	(6.05)
Intercept	-1.49	1.07	1.55	0.48
	(0.87)	(1.13)	(1.28)	(1.26)
Weighted mean task Δ	-1.48	3.70	-3.95	-4.49
D. Within industry: Some college				
Δ Computer use	7.95	18.14	-15.68	-17.77
1984–1997	(5.03)	(5.54)	(5.27)	(5.61)
Intercept	-1.88	-0.58	0.35	1.39
	(1.05)	(1.15)	(1.10)	(1.17)
Weighted mean task Δ	-0.33	2.96	-2.71	-2.08
E. Within industry: College graduates				
Δ Computer use	1.61	5.57	-0.78	-4.46
1984–1997	(3.42)	(3.35)	(4.85)	(5.70)
Intercept	0.25	0.10	-0.96	-0.12
	(0.71)	(0.70)	(1.01)	(1.19)
Weighted mean task Δ	0.57	2.22	-1.48	-1.98
F. Decomposition into within and between education group components				
Explained task Δ	2.52	3.11	-3.09	-2.79
Within educ groups (%)	23.7	77.9	91.7	111.1
Between educ groups (%)	76.3	22.1	8.3	-11.1

n in panels A–E is 140, 139, 140, 140, and 139 consistent SIC industries. Standard errors are in parentheses. Each column of panels A–E presents a separate OLS regression of ten times the annual change in industry-level task input for the relevant education group (measured in centiles of the 1960 task distribution) during 1980–1998 on the annual percentage point change in industry computer use during 1984–1997 (weighted mean 0.198) and a constant. Estimates are weighted by mean industry share of total employment (in FTEs) in 1980 and 1988. Estimates with no employment in the relevant educational category in 1980 or 1988 are marked as “.”

TABLE VI
COMPUTERIZATION AND CHANGES IN JOB TASK CONTENT WITHIN OCCUPATIONS 1977–1991
DEPENDENT VARIABLE: $10 \times$ ANNUAL WITHIN-OCCUPATION CHANGE IN QUANTILE OF TASK MEASURE,
MEASURED IN PERCENTILES OF 1984 TASK DISTRIBUTION

	A. Δ Nonroutine analytic			B. Δ Nonroutine interactive			C. Δ Routine cognitive			D. Δ Routine manual		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Computer use	2.94	3.57	4.02	5.70	5.86	7.08	-18.18	-16.56	-18.48	1.74	0.83	0.37
84–1997	(1.84)	(1.92)	(2.06)	(1.88)	(1.97)	(2.11)	(3.29)	(3.41)	(3.65)	(2.89)	(3.01)	(3.23)
College grad emp.	-4.79	-4.83	-4.47	-4.58	22.59	22.76	22.59	22.76	22.76	-16.07	-16.03	-16.03
84–1997	(5.54)	(5.54)	(5.68)	(5.67)	(9.86)	(9.85)	(9.86)	(9.85)	(9.85)	(8.70)	(8.71)	(8.71)
HS grad emp.	2.83	3.09	-0.19	0.52	16.97	15.86	16.97	15.86	15.86	-10.42	-10.70	-10.70
84–1997	(3.78)	(3.81)	(3.88)	(3.90)	(6.73)	(6.77)	(6.73)	(6.77)	(6.77)	(5.94)	(5.99)	(5.99)
Female emp.	-2.37	-2.37	-6.47	-6.47	10.14	10.14	10.14	10.14	10.14	2.47	2.47	2.47
84–1997	(3.94)	(3.94)	(4.03)	(4.03)	(6.99)	(6.99)	(6.99)	(6.99)	(6.99)	(6.19)	(6.19)	(6.19)
Intercept	-0.92	-0.91	-0.95	-0.46	-0.42	-0.52	0.56	0.14	0.30	0.42	0.70	0.74
	(0.40)	(0.41)	(0.41)	(0.41)	(0.42)	(0.42)	(0.71)	(0.72)	(0.73)	(0.63)	(0.64)	(0.64)
Weighted mean Δ	0.01	0.01	0.01	0.02	0.02	0.03	0.06	0.08	0.08	0.00	0.01	0.01
	(0.39)	(0.39)	(0.39)	(0.39)	(0.39)	(0.39)	(0.39)	(0.39)	(0.39)	(0.39)	(0.39)	(0.39)

n is 470 consistent three-digit Census Occupation Code (COC) occupations. Standard errors are in parentheses. Each column presents a separate OLS regression of ten times the annual change in the occupational task measure (measured in centiles of the 1984 distribution) between the 1977 and 1991 DOT revisions. Computer use, college graduate, high school graduate, and female employment shares are measured as ten times the annual change in the relevant measure from the 1984 and 1997 CPS. Weighted means are 0.183, 0.017, 0.05, and 0.017, respectively. Omitted education categories are some college and high school dropout. Estimates are weighted by the average occupational share of U. S. employment in 1984 and 1997. See Table I and Appendix I for definitions and examples of task variables.

(a) Table V: computerization and task input by education group.

(b) Table VI: task-content change within occupations.

Within-group and within-occupation evidence. These tables show that task change occurs inside education and occupation cells.

6.5 Table VII: Shifts in College-Equivalent Labor Demand Implied by Changes in Job Tasks, 1970–1998

Read:

- Table VII converts task changes into implied demand shifts for college-equivalent labor.
- Observed task shifts imply large increases in demand for college labor between 1970 and 1998.
- Computerization-predicted task changes account for a substantial share of the standard CES-measured demand shift.
- The table separates extensive-margin task shifts, intensive-margin task shifts, and combined extensive-plus-intensive changes.

Economic message: This table closes the loop from computerization to tasks to education demand. It connects the task model to rising wage inequality and the college premium. The paper argues that task

change can explain roughly 60 percent of the estimated relative demand shift favoring college labor from 1970 to 1998. The table also shows that within-occupation task changes contribute meaningfully, not just employment reallocation across occupations.

TABLE VII
SHIFTS IN COLLEGE-EQUIVALENT LABOR DEMAND IMPLIED
BY CHANGES IN JOB TASKS, 1970–1998

	1. 1970– 1980 extensive margin	2. 1980– 1990 extensive margin	3. 1990– 1998 extensive margin	4. 1970– 1998 extensive margin	5. 1980– 1998 extensive margin	6. 1980– 1998 intensive margin	7. 1980– 1998 extensive + intensive
A. $10 \times$ observed annual changes in DOT task measures (percentile changes relative to 1960 task distribution)							
Nonroutine analytic	3.02	2.97	3.12	3.04	3.05	−0.39	2.67
Nonroutine interactive	4.68	5.31	4.48	4.84	4.85	0.58	5.43
Routine cognitive	−0.14	−3.48	−4.88	−3.03	−4.26	−2.76	−7.02
Routine manual	1.63	−1.47	−3.88	−1.44	−2.81	0.74	−2.07
B. $10 \times$ predicted annual changes in DOT task measures (percentile changes relative to 1960 task distribution)							
	NIPA computer input measure			CPS computer use measure			
Nonroutine analytic	0.84	1.35	2.30	1.55	2.56	0.54	3.10
Nonroutine interactive	1.47	2.36	4.01	2.70	3.16	1.04	4.20
Routine cognitive	−1.05	−1.68	−2.86	−1.92	−3.14	−3.32	−6.46
Routine manual	−1.15	−1.86	−3.15	−2.12	−2.84	0.32	−2.52
C. $10 \times$ predicted annual changes in college-equivalent share of employment in percentage points, due to observed task shifts (panel A)							
Nonroutine tasks	1.53	1.40	1.63	1.51	1.53	−0.36	0.83
Routine tasks	−0.20	0.66	1.17	0.50	0.97	0.27	0.87
All tasks	1.33	2.06	2.80	2.01	2.49	−0.09	2.40

Panel A: Observed extensive margin task shifts are defined as the change in economywide input of each task (in percentiles of the 1960 task distribution) estimated using DOT 1977 occupational task measures applied to Census and CPS samples for 1970 to 1998 and summarized in Table II. Intensive margin shifts are measured as change in mean of DOT occupational task input (measured in centiles of the 1977 task distribution) between 1977 and 1991 DOT revisions, using the 1980 and 1998 occupational distributions of employment from the CPS MORG samples. See Table I and Appendix 1 for definitions and examples of task variables.

Panel B: Predicted task changes are calculated as the weighted mean of the NIPA computer investment measure or CPS computer use measure (as noted) multiplied by the coefficient from a regression of changes in industry or occupation task input on the relevant computer measure. NIPA coefficient estimates correspond to specification (1) of Table IV. CPS extensive margin computer task estimates correspond to Panel A of Table V. CPS intensive margin computer task estimates correspond to specification (1) of Table VI. (A nontrivial information issue is involved.)

(a) Table VII, part 1: observed and predicted task shifts.

TABLE VII
(CONTINUED)

	1. 1970– 1980 extensive margin	2. 1980– 1990 extensive margin	3. 1990– 1998 extensive margin	4. 1970– 1998 extensive margin	5. 1980– 1998 extensive margin	6. 1980– 1998 intensive margin	7. 1980– 1998 extensive + intensive
D. $10 \times$ annual changes in college-equivalent share of employment in percentage points, predicted by impact of computerization on task input (panel B)							
	NIPA computer investment measure			CPS computer use measure			
Nonroutine tasks	0.40	0.65	1.10	0.69	1.41	0.40	1.81
Routine tasks	0.29	0.48	0.81	0.51	0.80	1.04	1.84
All tasks	0.70	1.12	1.91	1.19	2.21	1.44	3.65
E. Estimated log demand shifts for college-equivalent/noncollege-equivalent labor 1970–1998 ($100 \times$ annual log changes)							
Using constant-elasticity of substitution model to estimate changes in college demand							
$\sigma = 0.0$	4.99	2.53	2.25	3.33			2.41
$\sigma = 1.4$	3.95	4.65	2.76	3.86			3.81
$\sigma = 2.0$	3.50	5.56	2.98	4.09			4.41
Using task model to predict changes in college demand							
Total task Δ (panel C)	1.23	1.29	1.43	1.31	1.56	−0.06	1.51
Predicted by computerization (panel D)	0.64	0.70	0.98	0.76	1.39	0.91	2.29

Panels C and D: Implied employment share changes for college-equivalent labor (in percentage points) are calculated as the inner product of observed or predicted changes in task input from panels A and B and the coefficient vector from a fixed coefficient model of educational input. For extensive margin task shifts, this coefficient vector is estimated from a regression of college equivalent employment (in FTEs) in 140 consistent CIC industries on the five DOT measures of industry task input (in centiles) and a constant using the 1980 MORG sample. For intensive margin task shifts, the coefficient vector is estimated from a regression of college equivalent employment in 470 COC occupations on the five DOT measures of occupational task input (in centiles) and a constant using the 1984 CPS sample. College-equivalents labor is defined as all workers with college or greater education plus half of those with some college.

Panel E: Fixed coefficients log relative demand shifts are calculated as the change in the log ratio of college-equivalent/noncollege-equivalent employment using the initial (1970, 1980, or 1990) college-equivalent/equivalent employment share in full-time equivalents and the implied percentage point change in this share from Panels C and D.

(b) Table VII, part 2: implied college-equivalent demand shifts.

Table VII: Task shifts and college-equivalent labor demand. The final table translates task changes into the language of the wage-inequality literature.

7 Short summary

- The paper asks what computers do and how that changes labor demand.
- Computers substitute for routine tasks that can be codified into explicit rules.
- Computers complement nonroutine analytic and interactive tasks.
- From 1960 to 1998, routine task inputs decline while nonroutine cognitive task inputs rise.
- Computerization predicts these task shifts across industries and occupations.
- The effects occur within industries, within occupations, and within education groups.
- The model explains a large share of the rising relative demand for college labor.
- The contribution is to replace a vague “skill-biased technical change” label with a testable task-based mechanism.

8 Conclusion

8.1 Core conclusion

Autor, Levy, and Murnane show that recent technological change is best understood as task-biased rather than simply skill-biased. Computers substitute for codifiable routine tasks and complement nonroutine problem-solving and communication tasks. This task-level mechanism explains why computerization is associated with increased demand for educated labor.

8.2 Economic intuition

The intuition is simple but powerful. Computers are good at executing rules. They are less good at flexible judgment, persuasion, hypothesis formation, and complex interaction. When computers become cheap, firms automate routine tasks and reorganize production around nonroutine cognitive tasks. Workers who are relatively good at these tasks become more valuable.

8.3 Incremental contribution revisited

The paper’s incremental contribution is the task framework. It provides a microfoundation for observed correlations between computers and education demand. It also generates empirical predictions about routine and nonroutine task inputs that can be tested in DOT, Census, CPS, and computer-use data.

8.4 Implications for wage inequality and polarization

The paper explains why computerization can raise the college wage premium: college workers are more concentrated in nonroutine cognitive tasks that computers complement. It also sets the stage for later job-polarization research: routine middle-skill tasks are vulnerable to automation, while abstract high-skill tasks and some manual service tasks are less directly substitutable.

8.5 Limitations

The evidence is not based on randomized computer adoption. DOT task measures are imperfect and may not capture all within-job changes. Other forces—trade, organizational change, product innovation, and broader capital deepening—may also affect task demand. But the consistency of the patterns across time, industries, occupations, and education groups makes the task interpretation compelling.

8.6 Takeaway

The central takeaway is that technology changes work by changing tasks. Understanding labor demand therefore requires measuring what workers do, not only their schooling level or occupation title. This insight became a foundation for modern research on automation, polarization, and routine-biased technological change.